

Experiment 4: Artificial Neural Network using Backpropagation

Aim

To implement an Artificial Neural Network (ANN) using the Backpropagation algorithm and test it using the Iris dataset.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the Iris dataset.

Step 3: Split the dataset into training and testing sets.

Step 4: Standardize the feature values using StandardScaler.

Step 5: Create an Artificial Neural Network using MLPClassifier.

Step 6: Train the model using the training data.

Step 7: Predict the class labels for the test data.

Step 8: Calculate and display the accuracy of the model.

Program

```
#Experiment 4: Artificial Neural Network using Backpropagation
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()

X_train, X_test, y_train, y_test = train_test_split(
    iris.data,
    iris.target,
    test_size=0.3,
    random_state=0
)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    max_iter=1000,
    random_state=0
)

model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
```

Output:

```
... Accuracy: 0.9777777777777777
```

Result

Thus, the Artificial Neural Network using the Backpropagation algorithm was successfully implemented, and the model was trained and tested using the Iris dataset with high classification accuracy.